A/B Test Analysis: Control (A) vs Treatment (B)

Prepared October 31, 2026 · Simulated purchase-conversion experiment · Beginner-friendly walkthrough

Confidence key: [Certain] = hard arithmetic or evidence from your numbers. [Likely] = strong inference. [Guessing] = filling gaps. Most of this document is [Certain]; the guesses are labelled and kept to a minimum.

Bottom line

- [Certain] Group A converts at 12.5% and Group B at 18.2%. Both figures check out against your funnel table.
- [Certain] B is 5.7 percentage points higher, a 45.6% relative lift.
- [Certain] Your chi-square statistic (12.067) and p-value (0.000513) agree with each other.
- [Certain] At the 0.05 threshold, the null hypothesis is rejected. The difference is statistically significant.
- [Likely] Group B performed better. The extra purchases come from earlier funnel steps (cart and checkout), not from the final purchase step.
- [Certain] One thing in your request is wrong: the table you called a "contingency table" is not one. It has no user counts, so the chi-square cannot be recomputed from it alone. Details in Section 1.

1. What you gave me, and what is missing

Your funnel table

Step	Group A	Group B
Website Visits	100.00	100.00
Product Views	69.60	73.50
Added to Cart	44.25	50.61
Started Checkout	56.82	68.01
Purchases	71.43	71.94

Problem 1: these are step-to-step rates, not counts. [Certain] If they were a running funnel of the same 100 visitors, "Added to Cart" (44.25) could not be followed by "Started Checkout" (56.82), because the number of people can only shrink as they move down the funnel. So each row must be the percentage of people from the *previous* step who moved forward:

- 69.60% of visitors viewed a product
- 44.25% of those viewers added to cart
- 56.82% of those cart-adders started checkout
- 71.43% of those checkout-starters purchased

Problem 2: no sample sizes were supplied. [Certain] A chi-square test needs counts (how many people bought and how many did not in each group). "100.00" in the Website Visits row is a base for percentages, not a headcount. I will not invent a sample size, so the chi-square in Section 4 is a *verification of your reported numbers*, not an independent test from raw data.

Problem 3: the label "Purchases 71.43 / 71.94" is easy to misread. [Certain] It does not mean 71% of visitors buy. It means about 71% of people who *started checkout* completed the purchase. Overall conversion is much lower (12.5% and 18.2%).

2. Conversion rates (calculated independently)

Overall conversion = multiply the step rates together (as decimals).

Group A

0.6960 × 0.4425 × 0.5682 × 0.7143

- 0.6960 × 0.4425 = 0.30798
- 0.30798 × 0.5682 = 0.17499
- 0.17499 × 0.7143 = **0.1250 → 12.50%**

Group B

0.7350 × 0.5061 × 0.6801 × 0.7194

- 0.7350 × 0.5061 = 0.37198
- 0.37198 × 0.6801 = 0.25299
- 0.25299 × 0.7194 = **0.1820 → 18.20%**

[Certain] Your reported rates, 12.5% for A and 18.2% for B, are correct. This also confirms that reading the table as step-to-step rates is the right interpretation, because the products land exactly on your figures.

3. The difference in percentage points

- Absolute difference: 18.2% − 12.5% = **5.7 percentage points (pp)**
- Relative lift: 5.7 ÷ 12.5 = **0.456 → 45.6% higher**

Beginner explanation: A percentage *point* is a straight subtraction of two percentages. A percentage *change* compares the gap to the starting value. B is 5.7 points above A, which means B's rate is about 1.46 times A's. Saying "B improved by 5.7%" would be wrong and misleading; say "5.7 percentage points" or "45.6% relative lift."

Where the lift comes from [Certain]

Funnel step	A	B	Gap (pp)	B ÷ A
Product Views	69.60	73.50	+3.90	1.056
Added to Cart	44.25	50.61	+6.36	1.144
Started Checkout	56.82	68.01	+11.19	1.197
Purchases	71.43	71.94	+0.51	1.007

Multiplying the last column (1.056 × 1.144 × 1.197 × 1.007) gives about 1.456, which matches the overall lift. The biggest drivers are Started Checkout and Added to Cart. The final purchase step is nearly identical between groups.

[Likely] Whatever B changed (design, offer, flow) is working on getting people into cart and checkout, not on closing the sale at the last step. The data does not say what B actually changed, so what to do next with this is [Guessing] until you review the treatment itself.

4. Chi-square test: verification

What it tests: whether the gap between A and B is larger than random chance would normally produce.

- Null hypothesis (H0): A and B have the same true purchase rate.
- Alternative hypothesis (H1): the rates differ.

Step 1: the p-value matches the statistic. [Certain] With 1 degree of freedom, the chi-square statistic equals a squared z-score. √12.0673 = 3.474. The two-sided p-value for z = 3.474 is about 0.000513, which is exactly what you reported. So the two numbers are internally consistent.

Step 2: which sample size produces your statistic? The statistic cannot be recomputed from your table alone, but I can test one candidate. [Likely] Your numbers are reproduced exactly if each group had **1,000 users**. That is a reverse-engineered inference, not something you told me, so treat it as unconfirmed. If your real sample size differs, tell me and the result will change.

Contingency table under that assumption

Group	Purchased	Did not purchase	Total
A	125	875	1,000
B	182	818	1,000
Total	307	1,693	2,000

Expected counts if H0 were true (same rate in both groups): purchased = 307 ÷ 2 = 153.5 per group; not purchased = 1,693 ÷ 2 = 846.5 per group.

Without continuity correction, $\chi^2 = \Sigma\ (\text{Observed} - \text{Expected})^2 \div \text{Expected}$:

- A purchased: $(125 - 153.5)^2 \div 153.5 = 5.2915$
- B purchased: $(182 - 153.5)^2 \div 153.5 = 5.2915$
- A not purchased: $(875 - 846.5)^2 \div 846.5 = 0.9595$
- B not purchased: $(818 - 846.5)^2 \div 846.5 = 0.9595$
- Total = **12.50** ($p \approx 0.0004$)

With Yates continuity correction (the default in many tools, including SciPy), subtract 0.5 from each $|O - E|$ before squaring, so each deviation becomes 28 instead of 28.5:

- Each purchased cell: $28^2 \div 153.5 = 5.1075$
- Each not-purchased cell: $28^2 \div 846.5 = 0.9262$
- Total = $2 \times 5.1075 + 2 \times 0.9262$ = **12.067**

[Certain] 12.067 is exactly your reported statistic. So your result most likely came from a 2×2 table with 125/1,000 vs 182/1,000 purchasers and the Yates correction.

Degrees of freedom: (rows − 1) × (columns − 1) = (2 − 1) × (2 − 1) = **1**. [Certain] This matches your report.

Critical value: at $\alpha = 0.05$ with 1 df, the cutoff is **3.841**. Your 12.067 is more than three times larger.

5. Interpreting the p-value (threshold 0.05)

- p = 0.000513, which is about 0.05%, or roughly 1 in 1,950.
- 0.000513 < 0.05, so the result is **statistically significant**. It is also below 0.01 and 0.001.

Beginner explanation: Pretend A and B are truly identical. The p-value is the chance of seeing a gap this large, or larger, just from random luck. Here that chance is about 0.05%, which is very small. So luck is a poor explanation for the gap.

What the p-value does NOT mean [Certain]

- It is not the probability that the null hypothesis is true.
- It is not the probability that B is better.
- It says nothing about whether the gap is large enough to matter commercially. Section 3 covers size.

6. Should the null hypothesis be rejected?

[Certain] **Yes.** The p-value (0.000513) is below 0.05 and the statistic (12.067) is above the 3.841 cutoff. There is statistically significant evidence that A and B have different purchase rates.

A correct way to phrase it: "We reject the null hypothesis of equal conversion rates." Avoid "we proved B is better." Tests give evidence, not proof.

7. Which group performed better?

[Certain] **Group B**, on both measures that matter: higher conversion (18.2% vs 12.5%) and a gap that is statistically significant.

Size of the gap, if n = 1,000 per group [Likely, conditional on that assumption]

- Standard error of the difference ≈ 1.61 pp
- 95% confidence interval for the gap ≈ 5.7 ± 3.15 → **about 2.5 to 8.9 pp**
- Even the low end of that range is a real improvement.

Caveats before shipping B

1. [Certain] The data is simulated. A real-world result needs a real experiment.
2. [Certain] Chi-square tests only the final outcome. It does not test each funnel step separately, so statements like "B is significantly better at checkout" need their own tests with counts per step.
3. [Likely] A valid test needs random assignment of users, independent users, and a sample size fixed in advance. I cannot verify any of these from your table.

4. [Likely] Statistical significance does not guarantee business value. Compare the lift against the cost of the change.

8. Is anything in your results incorrect?

Item	Verdict
Conversion A = 12.5%	Correct
Conversion B = 18.2%	Correct
Chi-square = 12.0673	Consistent with 125/1,000 vs 182/1,000 plus Yates correction (my inference of n)
p-value = 0.000513	Correct for that statistic at 1 df
df = 1	Correct
Threshold 0.05	Standard choice; fine
Table called a "contingency table"	**Incorrect label.** It is a step-to-step funnel-rate table with no counts
"Website Visits = 100"	Base for percentages, not a sample size
Purchases row (71.43 / 71.94)	Correct as step rates, but easily mistaken for overall conversion

What I would do instead: report the raw counts (users and purchasers per group) alongside the funnel rates, and quote the confidence interval for the difference, not only the p-value. **The risk in the current presentation** is that a reader assumes the table is a contingency table and either miscomputes the chi-square or treats 71% as overall conversion.

9. Quick glossary

- **Conversion rate:** purchasers ÷ total users in the group.
- **Percentage point:** the plain difference between two percentages.
- **Null hypothesis (H0):** the "nothing is different" assumption the test tries to disprove.
- **Chi-square (χ^2):** a measure of how far observed counts are from the counts expected if H0 were true.
- **Degrees of freedom:** for a 2×2 table, always 1.
- **p-value:** the chance of a gap this big or bigger from luck alone, assuming H0.

- **Yates correction:** a small adjustment that makes the test slightly more conservative for 2×2 tables.
- **Confidence interval:** a plausible range for the true gap.